SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1004	**Course Title:**	Software Engineering
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	**Total Marks:**	25
CO2 Statement:	*Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills*		
Student Name:	Bhuvaneshwari J	**Reg. No.:**	192572141
Date:	27.10.26	**Duration:**	60 minutes

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Scenario-Based Requirement Analysis

Instructions to Students:

Each student will be assigned an individual software project problem statement. Analyze the given scenario and prepare a complete Software Requirement Analysis document by following the steps below.

Question

Based on the **assigned problem statement**, perform a **Scenario-Based Requirement Analysis** and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

1. **Study and Analyze the Scenario**
   - Carefully understand the given problem statement.
   - Identify the business domain, stakeholders, existing challenges, and system objectives.
2. **Identify Functional and Non-Functional Requirements**
   - List all functional requirements required by the proposed system.
   - Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.
3. **Identify Actors and Use Cases**
   - Identify all primary and secondary actors interacting with the system.
   - List the major use cases associated with each actor.
   - Draw a Use Case Diagram representing the interactions between actors and the system.
4. **Prepare the Software Requirement Specification (SRS)**
   Develop an SRS document containing:
   - Introduction
   - Problem Description

- Scope of the System
- Functional Requirements
- Non-Functional Requirements
- Use Case Descriptions
- Data Requirements
- External Interface Requirements
- System Features

5. **Identify Assumptions and Constraints**
   - List the assumptions considered while developing the system.
   - Identify technical, operational, economic, legal, and time-related constraints affecting the project.
6. **Validate Requirement Completeness and Consistency**
   - Verify whether all stakeholder requirements have been addressed.
   - Check for ambiguity, redundancy, inconsistency, and missing requirements.
   - Suggest improvements to enhance the quality and completeness of the requirement specification.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis (30%)	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification (25%)	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modelling (20%)	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS)	SRS is well-structured, complete, professionally organized, and	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or

Documentation (15%)	includes all required sections.			lacks essential components.
Assumptions, Constraints, and Requirement Validation (10%)	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	**Mark**
Scenario Understanding and Problem Analysis (30%)	/5
Functional and Non-Functional Requirement Identification (25%)	/5
Actor Identification and Use Case Modelling (20%)	/5
Software Requirement Specification (SRS) Documentation (15%)	/5
Assumptions, Constraints, and Requirement Validation (10%)	/5
TOTAL	/25

Software Requirement Specification (SRS)
Smart Construction Project Monitoring System
SIMATS Engineering (Saveetha School of Engineering) | Department of Computer Science and Engineering
Course: CSA10 / CSA1004 – Software Engineering | CO2 – Assessment Tool 1: Scenario-Based Requirement Analysis | Problem Statement No. 29
Student Name: Bhuvaneshwari J | Reg. No.: 192572141 | Date: 27.10.26

1. Introduction

1.1 Purpose

This document is the Software Requirement Specification (SRS) for the Smart Construction Project Monitoring System (SCPMS). It describes what the system must do, the constraints under which it must operate, and how it interacts with its users and external systems. It is written for project managers, site engineers, procurement and finance teams, developers, testers and the client, and acts as the agreed baseline for design, development and acceptance testing.

1.3 Definitions and Abbreviations

In this document, SCPMS means the Smart Construction Project Monitoring System and SRS means Software Requirement Specification. BOQ (Bill of Quantities) is the itemised list of work and materials with their quantities and rates. A milestone is a significant checkpoint in the schedule that has a due date, and a Gantt chart is a bar chart that shows tasks against a timeline. EVM (Earned Value Management) is a technique to measure progress and cost together; SPI (Schedule Performance Index) and CPI (Cost Performance Index) are its two main ratios. RBAC means Role-Based Access Control, where each user can use only the functions allowed for his or her role. Requirements are numbered FR-xx (functional), NFR-xx (non-functional) and UC-xx (use cases), and priorities follow the MoSCoW scheme (Must, Should, Could).

2. Problem Description

2.1 Business Domain

The system belongs to the construction and project management domain. A construction project involves many parallel activities carried out by several contractors and sub-contractors, a large workforce, a continuous flow of materials, a fixed budget and a contractual deadline. Because all of these depend on each other, a problem in one area (for example a late cement delivery) quickly spreads to others (idle labour, a missed milestone, penalty charges).

2.2 Existing Challenges

In most sites today, information is held in paper registers, spreadsheets, WhatsApp messages and phone calls. This causes the following problems:

- Progress information reaches the project manager late and is often inaccurate, so delays are noticed only after they have grown.
- Material stock is not tracked against requirement, leading to shortages that stop work or over-ordering that blocks cash.

- Worker attendance, skills and wages are recorded manually, which creates errors, proxy attendance and poor allocation of labour.
- Expenses are compared with the budget only at month-end, so cost overruns are found too late.
- Risks and site issues (safety incidents, weather, design changes, equipment breakdown) are not logged or followed up systematically.
- Reports are prepared manually from many sources, which is slow and inconsistent.
- Coordination between contractors, engineers and management is weak because there is no single shared source of truth.

2.3 Stakeholders

The **Project Manager** is the main user and needs complete visibility of progress, delays, cost and risk in order to take decisions. The **Site Engineer or Supervisor** updates daily activity progress, attendance and material usage from the site. **Contractors and sub-contractors** view their assigned tasks, submit progress and raise material requests.

The **Procurement or Store Officer** maintains stock, purchase orders and supplier deliveries, while the **Finance or Accounts Officer** looks after the budget, expense approval and cost reports. The **Safety or Quality Officer** records risks, incidents and inspection results. The **Client or Owner** is interested in progress and milestone reports and in assurance that the project will finish on time. Finally, the **System Administrator** manages user accounts, configuration, backup and security.

2.4 System Objectives

1. Manage construction projects, their phases and activities.
2. Track project milestones against the planned schedule.
3. Monitor material availability and consumption.
4. Record workforce information and attendance.
5. Track project expenses against the budget.
6. Identify project delays automatically and early.
7. Manage risks and issues through to closure.
8. Generate progress reports for managers, clients and finance.

3. Scope of the System

3.1 In Scope

- Project, phase, task and milestone management with planned and actual dates and progress percentage.
- Material master, stock tracking, requisitions, purchase orders, supplier deliveries and consumption records.
- Worker and contractor registration, skill records, attendance and task allocation.
- Budget definition, expense recording, approval and variance analysis.
- Automatic delay detection, alerts and notifications.
- Risk and issue register with severity, owner, mitigation and status.
- Real-time dashboard and exportable reports (PDF/Excel).
- Role-based access for all stakeholders through a web application and a mobile-friendly site interface.

3.2 Out of Scope

- Architectural or structural design (CAD/BIM authoring).
- Full payroll processing and statutory tax filing (only wage data is exported).
- Online payment gateway handling.
- Heavy-equipment telematics hardware (can be a future extension).

4. Functional Requirements

The functional requirements are grouped by module. Unless stated otherwise, each requirement has Must priority; those marked (S) are Should and (C) are Could.

4.1 Project and Milestone Management

- FR-01: The system shall allow authorised users to create, edit, view and archive projects with name, location, client, budget, start date and end date.
- FR-02: The system shall divide a project into phases and tasks with planned dates, duration, dependencies and assigned contractor.
- FR-03: The system shall define milestones with due dates and show their status (Not Started, In Progress, Completed, Delayed).
- FR-04: The system shall let site engineers update task progress with percentage complete, actual dates, remarks and site photos.
- FR-05 (S): The system shall display the schedule as a Gantt chart comparing planned and actual progress.

4.2 Material Management

- FR-06: The system shall maintain a material master with item, unit, rate, supplier and minimum stock level.
- FR-07: The system shall record receipts, issues and consumption per task and update stock automatically.
- FR-08: The system shall alert the store officer when stock falls below the minimum level or is insufficient for upcoming tasks.
- FR-09 (S): The system shall support requisitions, approvals and purchase orders and track supplier delivery dates.
- FR-10 (C): The system shall record wastage and compare actual consumption with the BOQ estimate.

4.3 Workforce Management

- FR-11: The system shall register workers and contractors with ID, trade, wage rate and contact details.
- FR-12: The system shall record daily attendance (manual, QR or biometric) and working hours.
- FR-13 (S): The system shall allocate workers to tasks and warn about over-allocation or skill shortage.
- FR-14 (S): The system shall calculate labour cost from attendance and wage rate.

4.4 Expense and Budget Management

- FR-15: The system shall define the budget by project, phase and cost category (material, labour, equipment, overhead).

- FR-16: The system shall record expenses with date, category, vendor, amount and bill, and route them for approval.
- FR-17: The system shall compare actual cost with budget and compute variance, CPI and forecast cost at completion.
- FR-18: The system shall notify the manager when expenses cross a configurable percentage of the budget, such as 80% or 100%.

4.5 Delay Identification, Risks and Issues

- FR-19: The system shall compare planned and actual progress and flag delayed tasks and milestones automatically.
- FR-20 (S): The system shall compute SPI, estimate a revised completion date and send alerts by email, SMS or in-app notification.
- FR-21 (S): The system shall record the reason for each delay, such as weather, material shortage, labour shortage, approval or design change.
- FR-22: The system shall maintain a risk register with category, probability, impact, risk score, owner and mitigation plan.
- FR-23: The system shall let staff log issues and incidents, assign them and track them until closure.
- FR-24 (S): The system shall highlight high-severity open risks and overdue issues on the dashboard.

4.6 Reporting, Dashboard and Administration

- FR-25: The system shall provide a real-time dashboard of progress, milestones, stock, cost, delays and risks.
- FR-26: The system shall generate daily, weekly and monthly progress reports and export them as PDF and Excel.
- FR-27 (S): The system shall filter reports by project, date range, contractor and phase.
- FR-28: The system shall provide secure login and role-based access control.
- FR-29: The system shall let the administrator manage users, roles, master data and alert thresholds.
- FR-30 (S): The system shall keep an audit log of create, update, approve and delete actions.

5. Non-Functional Requirements

The quality attributes below describe how well the system must perform its functions.

- **Performance (NFR-01, NFR-02):** Pages shall load within 3 seconds for up to 500 concurrent users, a monthly report shall be generated within 10 seconds, and site updates shall appear on the manager's dashboard within 5 seconds.
- **Security (NFR-03, NFR-04):** Every user shall log in with unique credentials, passwords shall be stored as salted hashes, access shall be role-based, data shall travel over HTTPS/TLS, and sessions shall expire after 15 minutes of inactivity.
- **Reliability and availability (NFR-05, NFR-06):** Financial and stock transactions shall be atomic so that no data is lost; availability shall be at least 99.5% with daily backup and recovery within 4 hours.
- **Usability (NFR-07, NFR-14):** A new site engineer shall be able to record progress after one hour of training; the interface shall be simple, mobile friendly and readable under WCAG 2.1 AA contrast guidelines.

- **Scalability (NFR-08):** The system shall grow to 100 simultaneous projects and 10,000 workers on cloud infrastructure without redesign.
- **Maintainability and portability (NFR-09, NFR-10):** A modular layered design with documented APIs shall allow new modules to be added easily, and the system shall run on current browsers and on Android and iOS devices.
- **Offline support (NFR-11):** Site users shall enter data offline and the system shall synchronise it when connectivity returns, because sites often have poor networks.
- **Auditability and data integrity (NFR-12, NFR-13):** Every change to cost, stock and progress shall be traceable to a user and a time, and inputs shall be validated to prevent negative or duplicate entries.

6. Actors and Use Cases

6.1 Actors

The system has eight primary actors, who are human users, and three secondary actors, which are external systems. The primary actors are the Project Manager, Site Engineer or Supervisor, Contractor, Store or Procurement Officer, Finance Officer, Safety or Quality Officer, Client or Owner (read-only) and System Administrator. The secondary actors are the Notification Service (email and SMS) that delivers alerts, the Biometric or QR Attendance Device that supplies attendance data, and the Supplier or Accounting System that exchanges purchase and payment data.

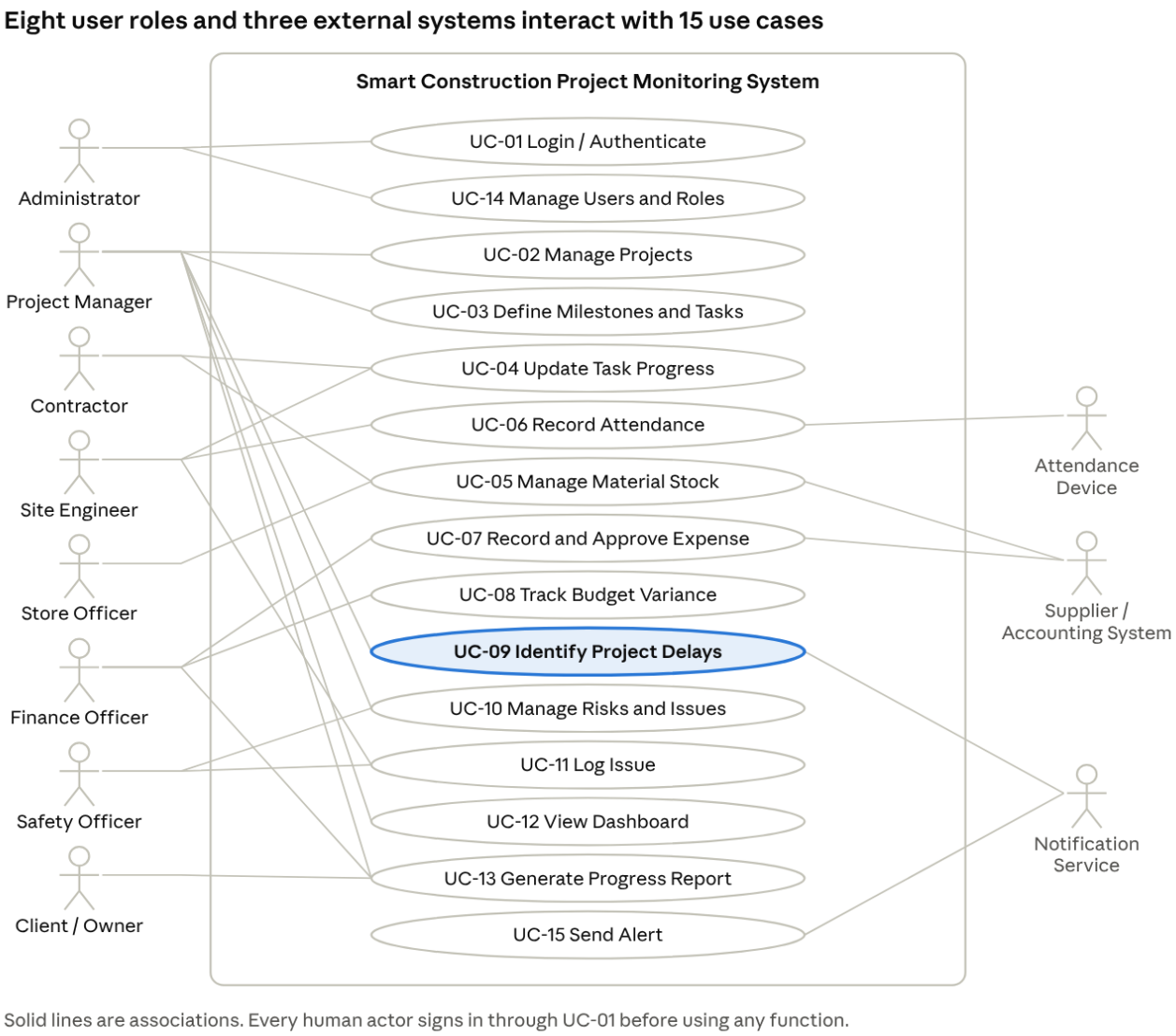
6.2 Major Use Cases per Actor

The Project Manager manages projects (UC-02), defines milestones and tasks (UC-03), identifies delays (UC-09), reviews risks, views the dashboard (UC-12) and generates progress reports (UC-13). The Site Engineer updates task progress (UC-04), records attendance (UC-06), records material usage (UC-05) and logs issues (UC-11). The Contractor views assigned tasks, submits progress (UC-04) and raises material requests (UC-05).

The Store Officer manages material stock (UC-05), creates purchase orders and receives deliveries. The Finance Officer records and approves expenses (UC-07), tracks budget variance (UC-08) and generates cost reports (UC-13). The Safety Officer manages the risk register (UC-10) and logs issues (UC-11). The Client views progress reports (UC-13). The Administrator handles login and authentication (UC-01), manages users and roles (UC-14) and configures alerts. The Notification Service sends alerts (UC-15) and the Attendance Device provides attendance data (UC-06).

6.3 Use Case Diagram

The use case diagram below shows the actors outside the system boundary and the main use cases inside it.



Human actors (left) and external systems (right) connect by solid lines to the use cases they take part in; the highlighted UC-09 is the core delay-detection function.

7. Use Case Descriptions

UC-04: Update Task Progress

Actor: Site Engineer (primary) and Contractor.

Description: records the daily progress of a construction task.

Pre-condition: the user is logged in and the task is assigned to the project.

Related requirements: FR-04, FR-19, NFR-02.

Main flow: The engineer selects the project and task after the daily site inspection. The system shows the current progress. The engineer enters the percentage complete, actual dates, remarks and photos. The system validates the data, saves the update, recalculates milestone and project progress and refreshes the dashboard.

Alternate flow: If the data is invalid (for example progress lower than before or a future date), the system shows an error and asks for correction. **Post-condition:** the progress is stored and the dashboard shows the latest status.

UC-05: Manage Material Stock

Actor: Store Officer (primary) and the Supplier system (secondary).

Description: records receipt, issue and consumption of materials and monitors availability.

Pre-condition: the material exists in the material master.

Related requirements: FR-06 to FR-09.

Main flow: The officer selects the material and the transaction type (receipt or issue), then enters quantity, date, task and delivery note. The system checks the available quantity, updates the stock and compares it with the minimum level and the upcoming requirement. If stock is low, it raises a shortage alert.

Alternate flow: If the issue quantity is more than the available stock, the system rejects the transaction and suggests raising a requisition.

Post-condition: the stock balance is updated and an alert is generated where needed.

UC-07: Record and Approve Expense

Actor: Finance Officer (primary) and Project Manager (approver).

Description: enters a project expense and compares it with the budget.

Pre-condition: a budget is defined for the project and category.

Related requirements: FR-15 to FR-18.

Main flow: The officer enters the expense details and uploads the bill. The system validates the mandatory fields and sends the expense to the manager. When the manager approves, the system adds the amount to actual cost and recalculates the variance. If a threshold is exceeded, a budget alert is sent.

Alternate flow: If the manager rejects the expense, the status is set to Rejected with a reason and the officer is notified.

Post-condition: the approved expense is part of actual cost and the variance is updated.

UC-09: Identify Project Delays

Actor: Project Manager (primary) and Notification Service (secondary).

Description: the system compares planned and actual progress to detect delays.

Pre-condition: tasks have planned dates and progress updates.

Trigger: a progress update or the scheduled daily check.

Related requirements: FR-19 to FR-21.

Main flow: The system compares the actual percentage with the planned percentage of each task and marks tasks that are behind schedule as Delayed. It calculates the SPI and a revised end date and sends an alert to the manager through the Notification Service. The manager views the delayed tasks and records the reason and corrective action.

Alternate flow: If progress data is insufficient, the task is flagged as Update Pending.

Post-condition: delays are visible on the dashboard with their reasons and actions.

8. Data Requirements

8.1 Main Data Entities

The system stores its data in a relational database with the following main entities.

- **User:** UserID, name, role, email, phone, password hash and status. A user performs actions on all other entities.
- **Project:** ProjectID, name, location, client, start and end date, budget and status. A project has many phases, milestones, expenses and risks.
- **Phase/Task and Milestone:** TaskID, planned and actual dates, percentage complete, contractor and status; MilestoneID, due date, completion date and status. Both belong to a project.
- **Material, Material Transaction and Purchase Order:** material name, unit, rate, minimum and current stock, supplier; receipts and issues linked to tasks; orders with expected delivery date.
- **Worker, Contractor/Supplier and Attendance:** worker ID, trade, wage rate and contractor; attendance with date, in-time, out-time and hours linked to a worker and a task.
- **Budget and Expense:** planned amount per category; expense with vendor, amount, date, bill file and approval status.
- **Risk/Issue:** description, category, probability, impact, score, owner, mitigation and status.
- **Alert and Audit Log:** alert type, message, target user and read status; audit entries with user, action, old and new value and timestamp.

8.2 Data Rules

- Every record shall have a unique primary key and creation/modification timestamps.
- Stock quantity and expense amount cannot be negative; planned end date cannot be earlier than planned start date.
- Progress percentage must be between 0 and 100 and shall not decrease without a recorded reason.
- Financial, attendance and stock records shall be retained for at least 7 years and shall not be physically deleted (soft delete only).
- A daily automatic backup shall be taken and stored in a separate location.

9. External Interface Requirements

9.1 User Interfaces

A responsive web interface with a role-based dashboard, menu navigation, data-entry forms, tables with search and filters, Gantt and bar/pie charts, and colour-coded status indicators (green – on track, amber – at risk, red – delayed). Site forms shall be short and touch friendly for use on phones and tablets.

9.2 Hardware Interfaces

PC/laptop and smartphone/tablet with a browser; optional QR scanner or biometric device for attendance; camera for uploading site photos; printer for reports.

9.3 Software Interfaces

The system uses a relational database (MySQL or PostgreSQL) for persistent storage and an email and SMS gateway to deliver delay, stock and budget alerts. It exchanges expense and payment data with the existing accounting or ERP software through a REST API, uses a report engine to produce PDF and Excel files, and stores bills, drawings and site photos in cloud storage.

10. System Features

The system offers eight main features.

Project and milestone tracking (FR-01 to FR-05) covers projects, tasks, milestones, progress updates and the Gantt chart.
Material monitoring (FR-06 to FR-10) gives live stock, consumption tracking, shortage alerts and purchase orders.
Workforce management (FR-11 to FR-14) handles worker records, attendance, allocation and labour cost.
Expense and cost control (FR-15 to FR-18) provides budget set-up, expense approval, variance and forecast.
The **delay detection engine** (FR-19 to FR-21) compares plan with actual and warns early.
Risk and issue management (FR-22 to FR-24) provides a risk register with scoring, ownership and closure tracking.
Dashboard and reporting (FR-25 to FR-27) gives a real-time summary and exportable reports.
Security and administration (FR-28 to FR-30) covers authentication, roles, audit trail and configuration.

11. Assumptions and Constraints

11.1 Assumptions

1. Site engineers and supervisors have access to a smartphone, tablet or computer on site.
2. Internet connectivity is available at least intermittently; offline entry will be synchronised later.
3. Project plans (schedule, BOQ and budget) are available before the project is entered into the system.
4. Users provide accurate and timely data; progress is updated at least once a day.
5. Contractors and suppliers are willing to use the system and share data.
6. Management defines delay and budget alert thresholds at the start of the project.
7. Users have basic computer literacy and receive short training.
8. Each project has at least one assigned project manager.

11.2 Constraints

Technical constraints: the system must be web-based and work on common browsers and mobile devices, must integrate with the existing accounting software through APIs, and must cope with limited bandwidth at remote sites.
Operational constraints: it must be usable by site staff with limited technical skill, data must be updated daily to stay reliable, and the system must run 24x7 with minimal downtime.
Economic constraints: development, hosting and training must stay within the approved budget, and open-source tools are preferred to reduce licence cost.
Legal constraints: the system must follow labour and construction safety regulations, the IT Act and data-protection requirements for personal and financial data, and contract terms for record retention.
Time and organisational constraints: a prototype is expected in about two months and the full release within six months, delivered in phases, and existing manual practices must be changed gradually because some user resistance is expected.

12. Validation of Requirements

12.1 Completeness Check

Every stakeholder objective is traced to at least one requirement. Managing projects is covered by FR-01, FR-02 and FR-28 to FR-30; milestone tracking by FR-03 to FR-05; material availability by FR-06 to FR-10; workforce information by FR-11 to FR-14; expense tracking by FR-15 to FR-18; delay identification by FR-19 to FR-21; risks and issues by FR-22 to FR-24; and progress reports by FR-25 to FR-27. The need for real-time information for managers is covered by FR-25 and NFR-02.
All eight objectives and all expected outcomes in the problem statement are mapped to at least one requirement, and every use case traces to a functional requirement.

12.2 Quality Check

The requirements were reviewed for ambiguity, redundancy, inconsistency, missing items, testability and feasibility.

- **Ambiguity:** words such as "real-time", "fast" and "delay" can be read differently. They were fixed by measurable definitions: real-time is 5 seconds (NFR-02), page load is 3 seconds (NFR-01), and a delay means actual progress is below planned progress beyond a configurable threshold.
- **Redundancy:** progress update appears in both the contractor and engineer flows, and report generation is repeated for finance and manager. These were merged into single use cases (UC-04, UC-13) with role-based access.
- **Inconsistency:** offline entry (NFR-11) conflicts with the 5-second rule (NFR-02). The rule now applies when online, and offline data is synchronised within one minute of reconnection.
- **Missing requirements:** the first list lacked an audit trail, data retention, offline support and an approval workflow, so FR-30, NFR-11, NFR-12 and the retention rule in section 8.2 were added.
- **Testability and feasibility:** vague statements were replaced with figures such as 99.5% availability, 500 users and 4-hour recovery, and biometric attendance and ERP integration were marked optional for phase 2 because they depend on hardware and vendor APIs.

12.3 Suggestions to Improve the Specification

- Review the SRS with all stakeholders (walkthrough/inspection) and obtain sign-off before design.
- Build a prototype of the dashboard and progress-entry screen to confirm usability early.
- Maintain a requirements traceability matrix (requirement to use case to test case) to control change.
- Add future enhancements: drone or IoT-based progress capture, AI-based delay prediction, BIM integration and a dedicated mobile application.
- Prioritise using MoSCoW so that Must-have requirements are delivered in the first release.

13. Conclusion

The Smart Construction Project Monitoring System replaces scattered manual tracking with a single integrated platform for projects, milestones, materials, workforce, expenses, delays and risks. This SRS has analysed the scenario, identified stakeholders and actors, defined 30 functional and 14 non-functional requirements, modelled the main use cases, specified data and interface needs, listed assumptions and constraints and validated the requirements for completeness and consistency. Implementing the specification will give project managers real-time visibility, enable early detection of delays, improve resource use and cost control and increase the probability of timely project completion